

Module 2 – CI/CD

1. CI/CD Introduction

What is CI/CD?

CI/CD stands for Continuous Integration and Continuous Deployment. It is a method to automate the process of building, testing, and deploying applications.

Why CI/CD Matters

- Faster development cycles
 - Reduced manual errors
 - Continuous delivery of features
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2. Jenkins

What is Jenkins?

Jenkins is an open-source automation tool used to build and manage CI/CD pipelines.

Key Features

- Automates build and deployment
- Supports plugins
- Easy integration with tools

Use Case

Automatically run tests and deploy code after every update

3. GitHub Actions

What is GitHub Actions?

GitHub Actions is a CI/CD tool that allows automation directly inside GitHub repositories.

Why Use It

- Easy setup
- Integrated with GitHub
- Automates workflows

Example Workflow:

```
name: CI Pipeline

on: [push]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - run: echo "Build Started"
```

4. Pipeline Flow

What is Pipeline?

A pipeline is a sequence of automated steps in software delivery.

Typical Flow

• Code → Build → Test → Deploy

Benefits

- Faster releases
 - Consistent process
 - Reduced human errors
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5. Automation

What is Automation?

Automation is the use of tools to perform tasks without manual intervention.

Why It Matters

- Saves time
- Reduces mistakes
- Increases efficiency

Examples

- Auto testing
 - Auto deployment
 - Scheduled jobs
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6. Key Concepts to Remember

- CI/CD automates development workflow
 - Jenkins is a popular CI tool
 - GitHub Actions simplifies automation
 - Pipelines define workflow steps
 - Automation improves speed and reliability
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7. Summary

CI/CD is a core DevOps practice that enables faster and more reliable software delivery. By using tools like Jenkins and GitHub Actions, teams can automate workflows, reduce errors, and deploy applications efficiently.